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ORIGINAL RESEARCH

Path tracking control of automated vehicles based on adaptive MPC in variable scenarios

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Abstract

For complex and dynamic high-speed driving scenarios, an adaptive model predictive control (MPC) controller is designed to ensure effective path tracking for automated vehicles. Firstly, in order to prevent model mismatch in the MPC controller, a tire cornering stiffness estimation algorithm is designed and a soft constraint on slip angle is added to further enhance the controller's precision in tracking trajectories and the vehicle's driving stability. Secondly, the improved particle swarm optimization (IPSO) method with dynamic weights and penalty functions is suggested to address the issue of insufficient accuracy in solving quadratic programming. Additionally, the standard particle swarm optimization (PSO) algorithm is used to seek the most suitable time horizon parameters offline to obtain the best time horizon data set under different vehicle speeds and adhesion coefficients, and then it is optimized online by an adaptive network-based fuzzy inference system (ANFIS) to enhance the model predictive controller's adaptability in different operating conditions. Finally, simulation experiments are conducted under three different operating conditions: docked roads, split roads, and variable vehicle speeds. The results indicate that the designed adaptive MPC controller can accurately and stably track the reference trajectory in various scenarios.

1 | INTRODUCTION

Automated driving can effectively reduce traffic accidents and congestion caused by human factors, thus enhancing driving safety and efficiency [1, 2], which has attracted increasing attention in recent years.

As the lowest-level module of automated driving, path tracking is the final step in achieving automated driving and a key factor in fully leveraging automated driving performance, which has emerged as a buzzword in automated driving research [3]. Now, the commonly used path tracking control algorithms of intelligent vehicles mainly include: PID control [4], sliding-mode control (SMC) [5], pure pursuit [6], Stanley control [7], linear quadratic regulator (LQR) [8], model predictive control (MPC) [9–11], and learning-based control [12]. Liu et al. [13] systematically analysed the model complexity, optimal performance, computational cost, and application scenarios of these algorithms. Among them, MPC is also known as rolling time-horizon optimal control, whose main control idea is to build a predictive model and solve the optimal control strategy with an optimization algorithm [14]. Since MPC is well suited to handle multi-constraint objective problems such as vehicle dynamics constraints and actuator saturation constraints, it has gradually dominated research on automated driving motion control. Chen et al. [15] designed two path tracking controllers based on curvature computation and MPC in complex unstructured environments, and the results show that as the speed increases, the MPC controller has better dynamic tracking performance due to its ability to better handle optimization problems with constraints. In [16], the MPC-HDDC hierarchical dynamic drifting controller is constructed, which enables integrated control of drifting and typical turning manoeuvres, dramatically improves the manoeuvring limits of automated vehicles, and has high tracking accuracy. Especially in recent years, with the continuous improvement of computer performance [17], MPC controllers has found extensive application in the realm of automated driving.

MPC has poor real-time performance due to the need for rolling optimization in each control cycle and generally re-quires corresponding simplification of the model. In [18], a linear MPC controller based on kinematics was designed, which features a simple structure and excellent real-time performance. However, under high-speed conditions, there is no guarantee that the vehicle can accurately and stably track the reference trajectory because the dynamic characteristics of the vehicle itself are not considered. Quirynen et al. [19] introduced a non-linear model predictive controller that considers vehicle dynamics, leading to a more accurate representation of the vehicle system's dynamic response. While it can enhance tracking accuracy, it also increases the computational burden of the controller, leading to poorer real-time performance. Wang et al. [20] designed a lateral and longitudinal integrated controller and verified that this algorithm could ensure that vehicles could complete high-speed lane changing tasks stably and efficiently in a multi-vehicle traffic environment. In [21], the yaw moment is integrated into the lateral and longitudinal control systems. The simulation results demonstrate that the designed controller can precisely follow the reference path and significantly enhance the vehicle's driving stability.

In order to enhance the adaptability of MPC controllers to complex environments, many scholars have begun to conduct relevant research on the dynamic parameters of MPC controllers. The PSO method is employed for offline optimization of the time horizon parameters of MPC controller, aiming to increase the adaptability of vehicles to the driving environment [22–24]. The fuzzy control algorithm has been integrated into conventional the model predictive controller to adaptively tune the weight coefficients of the cost function. Simulation experiments have shown that this not only enables the controller to have high tracking accuracy but also improves ride smoothness and comfort [25]. An MPC controller with variable sampling steps is designed by Guo et al. [26], and simulation experiments show that this variable sampling step control strategy can balance the requirements of tracking performance, lateral stability, and computation time. Zhang et al.[27] have developed a novel adaptive learning model predictive control (ALMPC) algorithm for trajectory tracking in automated ground vehicles (AGV) systems subject to parameter uncertainties and additive disturbances. Compared to RMPC, it reduces conservatism further while introducing a slight increase in computational complexity.

At present, research on MPC rarely considers the issue of parameter mismatch in predictive models, as well as the impact of road adhesion coefficient and vehicle velocity on the precision and stability of path tracking. Furthermore, in high-speed automated driving scenarios that are complex and dynamic, a path tracking controller with fixed time horizon parameters exhibits inadequate tracking accuracy and instability. This poses challenges in meeting the tracking requirements in varying scenes.

In light of the above analyses, the main contributions of this paper can be summarized as follows:

1) To solve the defects of MPC controllers with fixed time horizon parameters, an adaptive model predictive controller combined with PSO algorithm is designed. The effects of vehicle velocity and road adhesion coefficient on the controller are evaluated comprehensively, and the MPC controller are optimized offline by the PSO method and online by an ANFIS to adaptively adjust its time horizon parameters, which can improve the tracking capability of the MPC controller for changing vehicle velocity and variable adhesion on roads.
2) To improve the solving accuracy of MPC controllers, an IPSO algorithm with a penalty function is designed, which transforms multi-constraint programming problems into unconstrained programming problems and sets hot start based on the special form of the MPC solution vector. Compared with the traditional quadratic programming solution method, it is easier to get the optimal control rule, and the solving time of each cycle is short, with good real-time performance.
3) To prevent model parameter mismatch, a cornering stiffness estimator utilizing recursive least squares method (RLS) with a forgetting factor is designed, and a slip angle constraint is added to limit tire force to saturation.

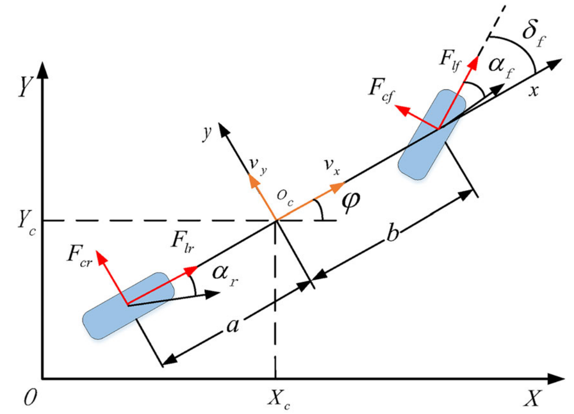

FIGURE 1 Vehicle monorail model.

2 | VEHICLE DYNAMICS MODEL

2.1 | Vehicle monorail model

A precise vehicle dynamics model can represent the vehicle's driving state better, making the tracking error smaller. However, vehicles are extremely complicated systems with multiple degrees of freedom, making them challenging to model [28], and employing a complex vehicle model will to some extent sacrifice the real-time responsiveness of the system. Therefore, comprehensively considering the demands for both system tracking precision and real-time responsiveness, a dynamics model with three degrees of freedom containing lateral, longitudinal, and yaw motions is developed, as illustrated in Figure 1 [29]. Here, xoy denotes the vehicle coordinate system, while XOY represents the global coordinate system. Moreover, the following assumptions are made [30]:

- The vehicle's centre of gravity is situated at the origin of the vehicle coordinate system.
- The roll, pitch, and vertical effects of the vehicle are neglected, and just longitudinal, lateral, and yaw movements are considered.

- The impact of suspension on the vehicle body is ignored.
- The vehicle is steered by the front wheels, and the cornering characteristics of the tires on both sides are identical.

Utilizing Newton's second law, the vehicle dynamics expressions for motion along the x-axis, y-axis, and rotation around the z-axis can be derived as follow [31]:

$$\begin{cases} m\ddot{x} = 2F_{lr} + m\dot{y}\dot{\varphi} + 2(F_{lf}\cos\delta_f - F_{cf}\sin\delta_f) \\ m\ddot{y} = 2F_{cr} - m\dot{x}\dot{\varphi} + 2(F_{lf}\sin\delta_f - F_{cf}\cos\delta_f) \\ I_z\ddot{\varphi} = 2a(F_{lf}\sin\delta_f + F_{cf}\cos\delta_f) - 2bF_{cr} \end{cases} \tag{1}$$

where m is the total vehicle mass; δ_f denotes the front wheel steering angle; I_z represents the vehicle yaw mass moment of inertia; φ is the vehicle's heading angle; $\dot{\varphi}$ denotes the vehicle's yaw rate; F_{lf} and F_{lr} are the longitudinal forces of the front and rear tires, respectively; F_{cf} and F_{cr} are the lateral forces of the front and rear tires, respectively; a and b are the distances from the centre of gravity to the front and rear axles, respectively; $\dot{x}$ and $\dot{y}$ represent the vehicle's longitudinal and lateral velocities, respectively; and $\ddot{x}$ and $\ddot{y}$ represent the vehicle's longitudinal and lateral accelerations, respectively.

Since the planned path by the planning layer is based on the global coordinate system, it is necessary to transform the velocity from the vehicle coordinate system to the global coordinate system [32]:

$$\begin{cases} \dot{X} = \dot{x}\cos\varphi - \dot{y}\sin\varphi \\ \dot{Y} = \dot{x}\sin\varphi + \dot{y}\cos\varphi \end{cases} \tag{2}$$

2.2 | Tire model

Tires are crucial components responsible for directly driving the vehicle and their interaction with the road surface leads to the generation of tire forces, which are essential for analysing vehicle dynamic characteristics. Therefore, the tire model established must precisely reflect the tire's dynamic properties. In this chapter, the extensively utilized "Magic Formula Tire" model from the fields of vehicle dynamics and tire mechanics is adopted [33].

Although the non-linear tire model has higher fitting accuracy, the expression is too complex, and the automated driving system requires high real-time performance at the control level. From Figure 2, it is evident that when the slip angle and slip rate are small, the tire model can be further simplified into a linear tire model as follows:

$$\begin{cases} F_{cf} = C_{cf}\alpha_f = C_{cf}\left(\delta_f - \dfrac{\dot{y} + a\dot{\varphi}}{\dot{x}}\right) \\ F_{cr} = C_{cr}\alpha_r = C_{cr}\dfrac{b\dot{\varphi} - \dot{y}}{\dot{x}} \\ F_{lf} = C_{lf}s_f \\ F_{lr} = C_{lr}s_r \end{cases} \tag{3}$$

The non-linear three-degree-of-freedom vehicle dynamics model can be derived from (1), (2), and (3), as shown in (4). And $\xi = [\dot{y}\dot{x}\varphi\dot{\varphi}YX]$ is chosen as the state variable, and $u = \delta_f$ [34] is the control variable.

$$\begin{cases} m\ddot{y} = 2\left[C_{cf}\left(\delta_f - \dfrac{\dot{y} + a\dot{\varphi}}{\dot{x}}\right) + C_{cr}\dfrac{b\dot{\varphi} - \dot{y}}{\dot{x}}\right] - m\dot{x}\dot{\varphi} \\ m\ddot{x} = 2\left[C_{lf}s_f - C_{cf}\left(\delta_f - \dfrac{\dot{y} + \dot{a}}{\dot{x}}\right)\delta_f + C_{lr}s_r\right] + m\dot{y}\dot{\varphi} \\ I_z\ddot{\varphi} = 2\left[aC_{cf}\left(\delta_f - \dfrac{\dot{y} + a\dot{\varphi}}{\dot{x}}\right) - bC_{cr}\dfrac{b\dot{\varphi} - \dot{y}}{\dot{x}}\right] \\ \dot{X} = \dot{x}\cos\varphi - \dot{y}\sin\varphi \\ \dot{Y} = \dot{x}\sin\varphi + \dot{y}\cos\varphi \end{cases} \tag{4}$$

3 | DESIGN OF MODEL PREDICTIVE LATERAL TRACKING CONTROLLER

Indeed, in complex obstacle avoidance scenarios involving low-adhesion, high-speed, and high-curvature conditions, the kinematic model may possess simplicity and high computational efficiency. However, it lacks consideration of the vehicle's own dynamic characteristics, leading to inadequate tracking performance under high-speed and low-adhesion conditions, leading to significant tracking errors and poor lateral stability. Therefore, the adoption of the established three-degree-of-freedom non-linear dynamic model is necessary to address these challenges effectively.

3.1 | Predictive model

The predictive model serves as the basis for MPC, and the established three-degree-of-freedom dynamic model is adopted as the predictive model for MPC in this chapter. To facilitate the derivation, (4) is expressed in the state-space form [21]:

$$\begin{cases} \dot{\xi}(t) = f(\xi(t), \mu(t)) \\ \eta(t) = C\xi(t) \end{cases} \tag{5}$$

In this model, $f(\cdot,\cdot)$ represents the state transfer matrix, $\xi(t)$ corresponds to the state variable, $\eta(t)$ denotes the output variable, and $\mu(t)$ refers to the control variable.

Since the above state-space equations are non-linear and non-linear model predictive control (NMPC) [35] cannot meet the system's real-time requirements, it is necessary to linearize the non-linear system. At present, linearization processing mainly includes two methods: exact linearization and approximate linearization. The exact linearization process is complex and has high accuracy but poor applicability; the approximate linearization method is simple and has low accu-

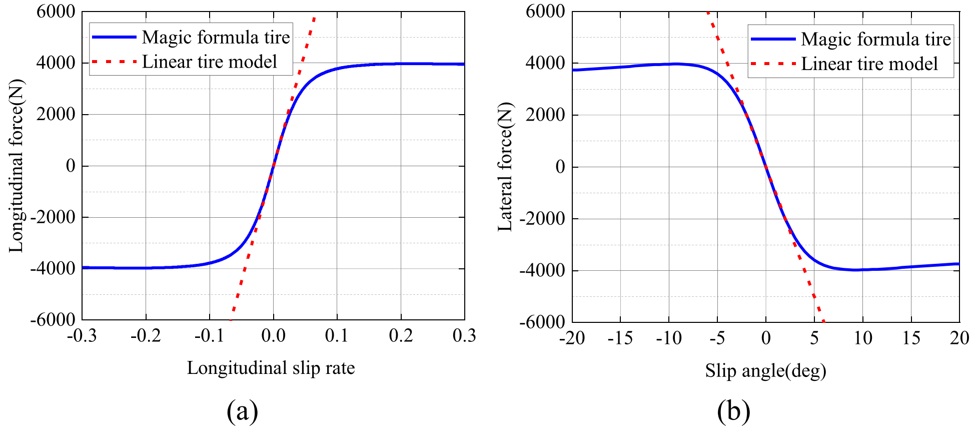

FIGURE 2 (a) The relationship between longitudinal force and slip rate, (b) the relationship curve between lateral force and slip angle.

racy but wide adaptability. Considering that the MPC has the function of feedback correction, which can continuously reduce the modelling error during operation, the approximate linearization method is used to linearize the vehicle non-linear system.

Using zero-order hold (ZOH), the system operating point $(\hat{\xi}_0(t), \hat{\mu}_0(t))$ is obtained. A second-order Taylor series approximation is utilized at the reference point and the terms beyond the second-order are neglected to obtain:

$$\dot{\xi}(t) = A(t)\xi(t) + B(t)u(t) \tag{6}$$

$$A(t) = \frac{\partial f}{\partial \xi}\Big|_{\hat{\xi}_0(t),\hat{\mu}_0(t)}, B(t) = \frac{\partial f}{\partial \mu}\Big|_{\hat{\xi}_0(t),\hat{\mu}_0(t)}$$

For ease of computer processing, the forward Euler method is adopted for the discretization of (6):

$$\xi(k+1) = A(k)\xi(k) + B(k)u(k) + d(k) \tag{7}$$

$$d(k) = \hat{\xi}_0(k+1) - A(k)\,\hat{\xi}_0(k) - B(k)u_0(k)$$

where $A(k) = I + TA(t), B(k) = TB(t)$, T represents the sampling time.

To accomplish smoother control of the underlying actuators, the control variables of the system are converted into control increments, and new state variables and equations are constructed as follows:

New state variables:

$$\tilde{\xi}(k) = \begin{bmatrix} \xi(k) \\ u(k-1) \end{bmatrix} \tag{8}$$

Updated state equations:

$$\begin{cases} \tilde{\xi}(k+1) = \tilde{A}_k\tilde{\xi}(k) + \tilde{B}_k\Delta u(k) + \tilde{d}_k \\ \eta(k) = \tilde{C}\tilde{\xi}(k) \end{cases} \tag{9}$$

where

$$\tilde{A}_k = \begin{bmatrix} A(k)B(k) \\ 0_{m\times n}I_m \end{bmatrix}, \tilde{B}_k = \begin{bmatrix} B(k) \\ I_m \end{bmatrix}, \tilde{d}_k = \begin{bmatrix} d(k) \\ 0_m \end{bmatrix}$$

$$\Delta u(k) = \mu(k) - \mu(k-1).$$

According to (9), if the current state variables and the control increment for the next time step are known, it is possible to deduce the state variable for the subsequent time step, and then the predicted state variable in the time horizon can be derived.

3.2 | Cost function

Stable and accurate tracking of the expected path are two important indicators in the automated vehicles' path tracking control. On the one hand, the accuracy is reflected in the accumulation of errors between the vehicle's real position and the target position during its operation. On the other hand, the stability is highly related to the rate and amplitude of change in the front wheel steering angle, and frequent steering manoeuvres will lead to the deterioration of the vehicle's lateral stability. Therefore, considering both of these requirements, the following cost function has been designed:

$$J(k) = \sum_{i=1}^{N_p} \left\| \eta(k+i|k) - \eta_{ref}(k+i|k) \right\|_Q^2 + \sum_{i=1}^{N_c-1} \left\| \Delta u(k+i|k) \right\|_R^2 + \rho\varepsilon^2 \tag{10}$$

where Q, R, and ρ represent the respective weight matrices for the corresponding terms; N_p and N_c denote the prediction horizon and control horizon, respectively, with the condition $N_c \le N_p$; and ε represents the relaxation factor. In (10), the first term represents the accuracy of the controller in tracking the desired trajectory; its second term depicts the vehicle's driving stability; and the last term introduces a relaxation factor to ensure finding feasible solutions in each control cycle.

Since the cost function is an optimization problem involving a quadratic form with multiple constraints, it can be reformulated into the standard form of quadratic programming for further analysis and solution, as shown in (11). This allows for the utilization of various quadratic programming algorithms and techniques to find the optimal solution efficiently.

$$\min \frac{1}{2}x^T Hx + f^T x \tag{11}$$

$$s.t.\begin{cases} A \cdot x \leq b \\ A_{eq} \cdot x = b_{eq} \\ l_b \leq x \leq u_b \end{cases}$$

where f denotes the gradient matrix, H represents the Hessian matrix, and x is the variable to be solved.

3.3 | Constraint conditions

To guarantee that the vehicle can stably and accurately follow the target trajectory, it is vital to impose constraints on the system's control variables, state variables, and output variables. The constraints to be met are as follows:

$$\begin{cases} \eta_{h,\min} \leq \eta(k) \leq \eta_{h,\max} \\ \eta_{s,\min} - \varepsilon \leq \eta(k) \leq \eta_{s,\max} + \varepsilon \\ \Delta U_{\min} \leq \Delta U(k) \leq \Delta U_{\max} \\ U_{\min} \leq A\Delta U(k) + U(k) \leq U_{\max} \end{cases} \tag{12}$$

where $\eta_{h,\min}$ and $\eta_{h,\max}$ represent the hard constraint boundaries of the system output; $\eta_{s,\min}$ and $\eta_{s,\max}$ denote the soft constraint boundaries of the output with the addition of the relaxation factor; $\Delta U_{\min}$ and $\Delta U_{\max}$ stand for the minimum and maximum values of the incremental front wheel angle; $U_{\min}$ and $U_{\max}$ refer to the minimum and maximum values of the front wheel steering angle.

In addition, to tackle extreme operating conditions where tire forces reach saturation, leading to side slip and potential loss of vehicle control, the incorporation of a lateral slip angle soft constraint is introduced to enhance the vehicle's driving stability. From Figure 3, it is evident that the road friction coefficient has a significant impact on the tire's lateral force, and the extreme value of the tire's lateral force decreases sharply and the linear region shortens as the road friction coefficient decreases. This indicates the necessity of constraining the slip angle based on the road friction coefficient. By keeping the tire's lateral force within the linear region, it ensures that the vehicle maintains sufficient lateral force to sustain the required centripetal force for steering.

Given that the built vehicle dynamics model has not include the sideslip angle as a state variable, it cannot be directly constrained and controlled as part of the system. Therefore, it is necessary to convert the sideslip angle into selected state variables and control variables. The transformation relationship between them can be described as:

$$\begin{cases} \alpha_f = \dfrac{\dot{y} + a\dot{\varphi}}{\dot{x}} - \delta_f \\ \alpha_r = \dfrac{\dot{y} - b\dot{\varphi}}{\dot{x}} \end{cases} \tag{13}$$

Similarly, by linearizing and discretizing (13) at the reference operating point $(\hat{\xi}_0(t), \hat{\mu}_0(t-1))$, a discrete linearized model of the sideslip angle output is obtained:

$$\alpha(k) = E_\alpha(k)\xi(k) + Z_\alpha(k)u(k-1) + e(k) \tag{14}$$

where

$$E_\alpha(k) = \begin{bmatrix} \dfrac{1}{\dot{x}} & \dfrac{-(\dot{y}+a\dot{\varphi})}{\dot{x}^2} & 0 & \dfrac{a}{\dot{x}} & 0 & 0 \\ \dfrac{1}{\dot{x}} & \dfrac{-(\dot{y}-b\dot{\varphi})}{\dot{x}^2} & 0 & \dfrac{-b}{\dot{x}} & 0 & 0 \end{bmatrix}, Z_\alpha(k) = \begin{bmatrix} -1 \\ 0 \end{bmatrix},$$

$$e(k) = \hat{\alpha}_0(k) - E_\alpha(k)\,\hat{\xi}_0(k) - Z_\alpha(k)\,\hat{u}_0(k-1).$$

Similarly, based on the system's output of sideslip angle and input of front wheel steering angle at the current moment, the output of sideslip angle can be deduced for any moment within the predicted time horizon, as follows:

$$\Omega_\alpha(k) = \mathrm{M}_\alpha \xi(k|k) + \mathrm{H}_\alpha \Delta U(k) + \Lambda_\alpha E(k) \tag{15}$$

where

$$\mathrm{H}_\alpha = \begin{bmatrix} E_\alpha(k)\tilde{B}_k & 0 & 0 & 0 \\ E_\alpha(k)\tilde{A}_k\tilde{B}_k & E_\alpha(k)\tilde{B}_k & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ E_\alpha(k)\tilde{A}_k^{N_c-1}\tilde{B}_k & E_\alpha(k)\tilde{A}_k^{N_c-2}\tilde{B}_k & \cdots & E_\alpha(k)\tilde{B}_k \\ E_\alpha(k)\tilde{A}_k^{N_c}\tilde{B}_k & E_\alpha(k)\tilde{A}_k^{N_c-1}\tilde{B}_k & \cdots & E_\alpha(k)\tilde{A}_k\tilde{B}_k \\ \vdots & \vdots & \ddots & \vdots \\ E_\alpha(k)\tilde{A}_k^{N_p-1}\tilde{B}_k & E_\alpha(k)\tilde{A}_k^{N_p-2}\tilde{B}_k & \cdots & E_\alpha(k)\tilde{A}_k^{N_p-N_c}\tilde{B}_k \end{bmatrix}$$

$$\Omega_\alpha(k) = \begin{bmatrix} \alpha(k+1|k) \\ \alpha(k+2|k) \\ \vdots \\ \alpha(k+N_c|k) \\ \vdots \\ \alpha(k+N_p|k) \end{bmatrix} E(k) = \begin{bmatrix} e(k) \\ e(k+1) \\ \vdots \\ e(k+N_c) \\ \vdots \\ e(k+N_p) \end{bmatrix}$$

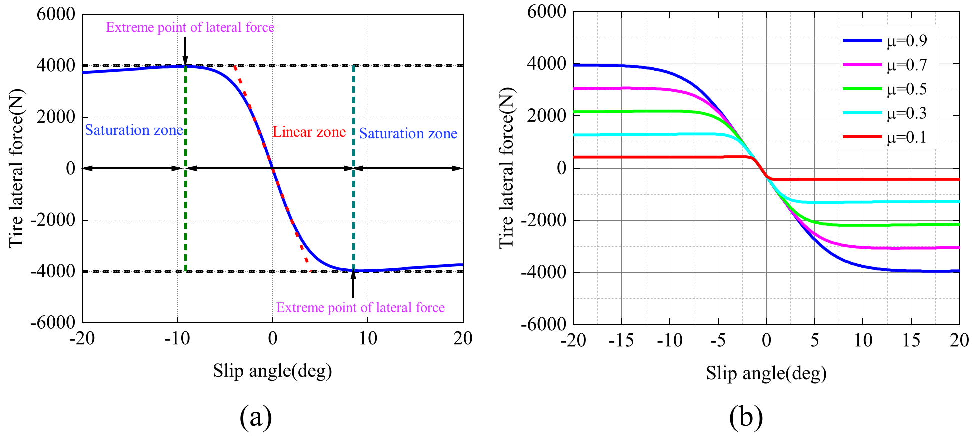

FIGURE 3 (a) Tire characteristic diagram, (b) tire lateral force under different adhesion coefficients.

$$\Lambda_\alpha = \begin{bmatrix} E_\alpha(k) & 0 & 0 & 0 \\ E_\alpha(k)\tilde{A}_k & E_\alpha(k) & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ E_\alpha(k)\tilde{A}_k^{N_c-1} & E_\alpha(k)\tilde{A}_k^{N_c-2} & \cdots & E_\alpha(k) \\ E_\alpha(k)\tilde{A}_k^{N_c} & E_\alpha(k)\tilde{A}_k^{N_c-1} & \cdots & E_\alpha(k)\tilde{A}_k \\ \vdots & \vdots & \ddots & \vdots \\ E_\alpha(k)\tilde{A}_k^{N_p-1} & E_\alpha(k)\tilde{A}_k^{N_p-2} & \cdots & E_\alpha(k)\tilde{A}_k^{N_p-N_c} \end{bmatrix},$$

$$\mathrm{M}_\alpha = \begin{bmatrix} E_\alpha(k)\tilde{A}_k \\ E_\alpha(k)\tilde{A}_k^{2} \\ \vdots \\ E_\alpha(k)\tilde{A}_k^{N_c} \\ \vdots \\ E_\alpha(k)\tilde{A}_k^{N_p} \end{bmatrix}.$$

The specific expression for the final sideslip angle constraint is as follows:

$$\begin{bmatrix} \mathrm{H}_\alpha \\ -\mathrm{H}_\alpha \end{bmatrix} \Delta U(k) \leq \begin{bmatrix} \alpha_{\max} \otimes ones_{N_c \times 1} - \mathrm{M}_\alpha \xi(k|k) \\ -\Lambda_\alpha D(k) - E(k) \\ -\alpha_{\min} \otimes one_{N_c \times 1} + \mathrm{M}_\alpha \xi(k|k) \\ +\Lambda_\alpha D(k) + E(k) \end{bmatrix} \tag{16}$$

4 | ESTIMATION OF TIRE CORNERING STIFFNESS

While the vehicle is moving, the tire cornering stiffness undergoes significant fluctuations due to variations in road surface adhesion coefficient and the shift of front and rear axle loads, which in turn will lead to a mismatch of the prediction model parameters of MPC and greatly affect the tracking accuracy of the controller. In this section, the real-time online estimation of tire cornering stiffness will be carried out using the RLS approach, aiming to further enhance the prediction precision of the MPC.

By sampling N sets of historical data, the linear regression equation between tire cornering force and slip angle is established according to (3) as shown below:

$$Y_{RLS}(k) = \alpha(k)\hat{\Phi}(k) + e(k) \tag{17}$$

where $Y_{RLS}(k) = [F_y(k)F_y(k-1)\cdots F_y(k-N+1)]$ represents the tire lateral force, $e(k)$ represents the deviation between the estimated and measured tire cornering stiffness, $\hat{\Phi}(k)$ symbolizes the tire cornering stiffness that is to be estimated, and $\alpha(k) = [\alpha(k)\alpha(k-1)\cdots\alpha(k-N+1)]$ stands for the tire slip angle.

For the front axle:

$$\alpha_f(k) = \frac{v_y(k) + a\dot{\varphi}(k)}{v_x(k)} - \delta_f(k) \tag{18}$$

For the rear axle:

$$\alpha_r(k) = \frac{v_y(k) - b\dot{\varphi}(k)}{v_x(k)} \tag{19}$$

To ensure that the estimated values from the linear regression equation closely approximate the measured values, the cost function is expressed in the following form:

$$\begin{aligned} J &= \min \sum_{k=1}^{N} \lambda^{N-k} e(k)^2 \\ &= \min \sum_{k=1}^{N} \lambda^{N-k} \left(Y_{RLS}(k) - \alpha(k)\hat{\Phi}(k)\right)^2 \end{aligned} \tag{20}$$

where λ denotes the forgetting factor, and its role is to decrease the contribution of historical data to prevent data saturation.

The partial derivatives of the above convex quadratic cost function for the parameters to be estimated are computed, and the partial derivatives are made to be zero, as shown in (21), to find the minimal value of the cost function.

$$\frac{\partial J}{\partial \hat{\Phi}} = -2\sum_{k=1}^{N} \lambda^{N-k}\alpha(k)(Y_{RLS}(k) - \alpha(k)\hat{\Phi}(k)) = 0 \quad (21)$$

Therefore, at the k-th step, the optimal parameter estimation $\hat{\Phi}(k)$ can be expressed as:

$$\hat{\Phi}(k) = \left[\sum_{k=1}^{N} \lambda^{N-k}\alpha^2(k)\right]^{-1} \left[\sum_{k=1}^{N} \lambda^{N-k}\alpha(k)Y_{RLS}(k)\right] \quad (22)$$

Let:

$$P(k) = \left[\sum_{k=1}^{N} \lambda^{N-k}\alpha^2(k)\right]^{-1}, \quad (23)$$

$$Q(k) = \left[\sum_{k=1}^{N} \lambda^{N-k}\alpha(k)Y_{RLS}(k)\right]. \quad (24)$$

As a result, (23) can be reformulated as:

$$\hat{\Phi}(k) = P(k)Q(k) \quad (25)$$

Further, (24) is transformed to:

$$P^{-1}(k) = \lambda P^{-1}(k-1) + \alpha(k)\alpha^T(k) \quad (26)$$

Multiply both sides of (26) by $P(k)$ and $P(k-1)$ simultaneously:

$$P(k-1) = \lambda P(k) + P(k)\alpha(k)\alpha^T(k)P(k-1) \quad (27)$$

Let the recursive gain be $K(k) = P(k)\alpha(k)$, so equation (27) can be rewritten as:

$$P(k) = \lambda^{-1}\left[P(k-1) - K(k)\alpha(k)P(k-1)\right] \quad (28)$$

The update formula for recursive gain can be obtained:

$$\begin{aligned} K(k) &= P(k-1)\alpha(k) \\ &\quad * \left[\lambda + Y_{RLS}(k) - \alpha^T(k)P(k-1)\alpha(k)\right]^{-1} \end{aligned} \quad (29)$$

Furthermore, the following recursive formula for cornering stiffness can be derived:

$$\hat{\Phi}(k) = \hat{\Phi}(k-1) + K(k)\left[Y_{RLS}(k) - \alpha(k)\hat{\Phi}(k-1)\right] \quad (30)$$

The optimal sequence of cornering stiffness estimates can be obtained by using N sets of historical data for recursion.

To confirm the efficacy of the introduced cornering stiffness estimation method, a joint simulation platform is constructed using MATLAB/Simulink and Carsim. And the forgetting factor λ is 0.98, the data length N is 100, 225/60 R18 tires are selected, and the steering wheel undergoes sinusoidal cycle steering, as depicted in Figure 4a.

Figures 4b, 4c and 4e respectively illustrate the effects of different road surface adhesion coefficients and the front and rear axle load transfer caused by changes in vehicle speed on tire cornering stiffness.

Figure 4b presents the estimation results of tire cornering stiffness on a high-adhesion road. At this point, the tire is operating within its linear range, and the algorithm responds quickly. Furthermore, the estimated cornering stiffness can ultimately fluctuate around a constant value with a small amplitude, which can be approximately equal to that constant value. Figure 4c demonstrates the estimated tire cornering stiffness on a low-adhesion road. At this stage, the tire has entered the non-linear region, and it can be clearly observed that there is a sharp decrease in the front and rear tire's cornering stiffness and the estimated value fluctuated greatly before the first 10 seconds. However, with the continuous accumulation of new data, the estimated cornering stiffness can ultimately stabilize around a constant value. Figures 4d and 4e, respectively, show the speed variation curve and the estimated results under variable speed conditions. Between 25 and 35 s, as the vehicle accelerates, there is a redistribution of loads between the front and rear axles, leading to a decrease in the estimated front tire's cornering stiffness and an increase in the estimated rear tire's cornering stiffness. But soon both the front and rear tires had entered a non-linear region, and the cornering stiffness of both tires starts to decrease, eventually regaining balance in the constant speed stage.

5 | OPTIMIZATION OF MODEL PREDICTIVE CONTROL BY PARTICLE SWARM OPTIMIZATION

Firstly, the control horizon N_c and the prediction horizon N_p in the MPC controller play an essential role in determining the accuracy and stability of path tracking. And the vehicle speed and road surface adhesion coefficient influence the selection of time horizon parameters. As the road surface adhesion coefficient decreases and the velocity of the vehicle increases, the prediction horizon should be extended accordingly, which can obtain path information from farther places so that the vehicle can steer in time. At the same time, the control horizon should be smaller to avoid frequent vehicle steering that may cause instability of the vehicle body. Therefore, there is an optimal pair of time horizon parameters in the MPC control system under certain road adhesion coefficients and vehicle speeds. Moreover, as the driving environment changes, the control parameters should be adjusted accordingly. However, the traditional manual tuning method is not only time-consuming and laborious

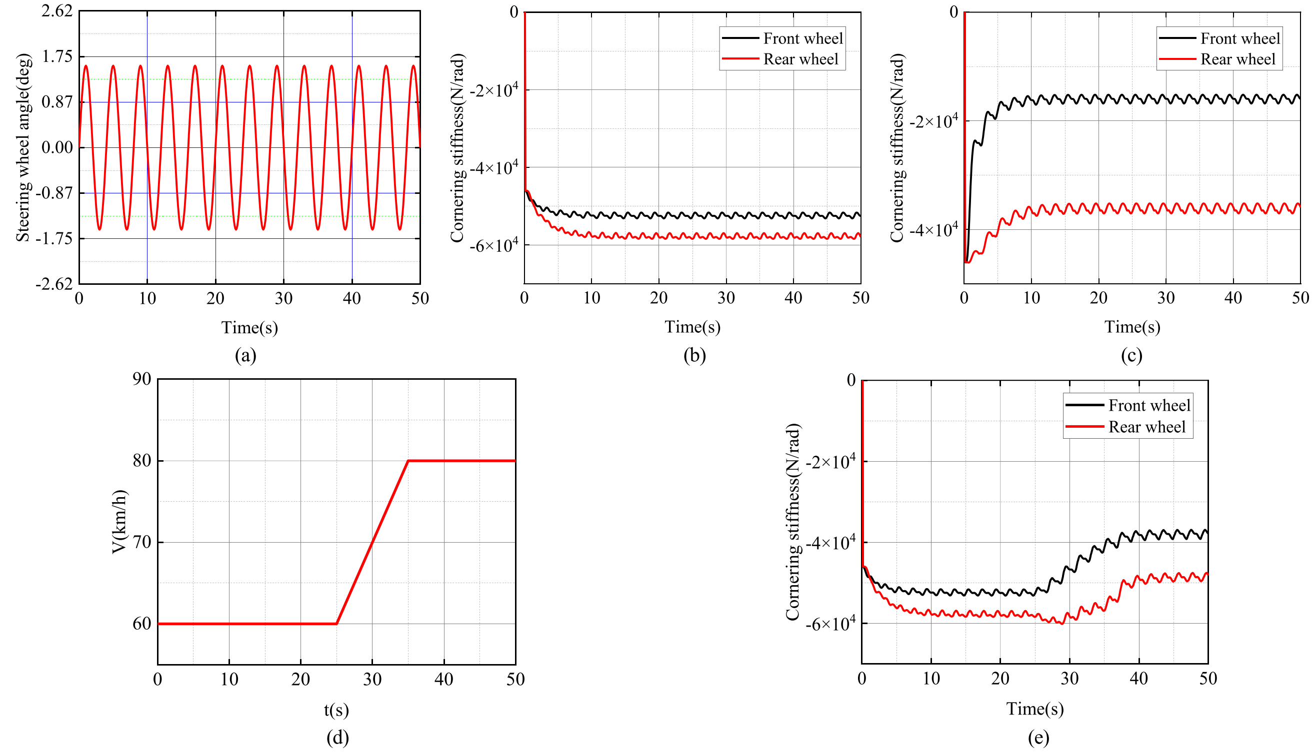

FIGURE 4 (a) Steering wheel angle change curve, (b) estimated cornering stiffness curve (speed: 60 km/h, road adhesion coefficient: 0.85), (c) estimated cornering stiffness curve (speed: 60 km/h, road adhesion coefficient: 0.3), (d) speed change curve, (e) estimated cornering stiffness curve (variable speed, road adhesion coefficient: 0.85).

but may also not be able to match the best parameters. Once the control parameters are determined, they cannot be adjusted during operation, which will increase the path tracking error and decrease stability. This chapter utilizes the PSO algorithm for offline optimization, aiming to determine the most suitable time horizon parameter pairs under diverse vehicle velocities and road surface adhesion coefficients, and then the offline data set is fitted by an ANFIS to achieve intelligent tuning under variable operating conditions. So, the more accurate the optimal time horizon dataset, the better the performance of the trained ANFIS, and the controller's adaptability to different driving environments improves.

Secondly, the core of the MPC control algorithm lies in solving a quadratic programming issue in every control cycle. However, traditional quadratic programming solution methods (e.g. effective set method, interior point method) are difficult to find the optimal solution and have insufficient accuracy when the matrix dimension is too high. In this chapter, an IPSO algorithm incorporating a penalty function is devised to solve the multi-constraint planning problem in the MPC control algorithm.

5.1 | Optimizing the time horizon parameters of MPC with PSO

To enhance the MPC controller's ability to adapt to different vehicle driving conditions, a standard PSO method is used to adaptively seek the best time horizon control parameters under the conditions of various road surface adhesion coefficients and vehicle velocities, and the lateral error ITAE indicator is used as the fitness function as described below:

$$J_{ITAE} = \int t\left|e(t)\right|dt \cdot e(t) = Y(i) - Y_{ref}(i) \tag{31}$$

where Y stands for the current location of the vehicle and Y_{ref} is the reference position of the desired trajectory.

The process of obtaining the optimal time horizon parameters for MPC through PSO involves the following four steps:

First step: Initialize the population with a population size of 15; set the maximum number of iterations to 20; particle position: $X = [N_P, N_c]$, upper boundary: $X_{\max}[40, 40]$, lower boundary: $X_{\min}[1, 1]$; particle velocity: $V = [v_{N_p}, v_{N_c}]$, upper boundary: $V_{\max}[1, 1]$, lower boundary: $V_{\min} = [-1, -1]$, and $c_1 = c_2 = 2, w = 0.6$.

Second step: Calculate the fitness value (ITAE index) and update the individual best P_{best} and global best G_{best} for the PSO algorithm.

Third step: Update the particle's speed and location.

$$\begin{aligned} V^{k+1} &= wV^k + c_1 r_1 (P_{best} - X^k) \\ &\quad + c_2 r_2 (G_{best} - X^k) X^{k+1} = X^k + V^{k+1} \end{aligned} \tag{32}$$

Fourth step: Meet the termination condition (maximum iteration times) and obtain the global optimal solution.

$$N_p = round(x_{N_p} + v_{N_p}), N_c = round(x_{N_c} + v_{N_c}). \quad (33)$$

5.2 | Optimizing QP solutions with IPSO

The IPSO algorithm and the standard PSO algorithm exhibit certain differences in solving optimization problems. First, standard PSO requires additional mechanisms to handle constraints effectively in MPC optimization. IPSO includes constraint-handling mechanisms, navigating constrained search spaces more effectively. Second, IPSO integrates problem-specific heuristics to enhance the optimization process by leveraging problem structure. Third, standard PSO typically relies on fixed parameters, which might not be suitable for all optimization problems. IPSO, however, introduces adaptive parameter tuning, allowing it to adjust parameters during the optimization process, thereby improving convergence. So, the IPSO algorithm converges faster and achieves higher solution accuracy.

Considering that traditional PSO algorithms are limited to solving unconstrained planning problems and that MPC in vehicle control requires real-time quadratic programming solutions with high precision and time-critical demands, using the IPSO algorithm to address the accuracy issues in quadratic programming becomes essential to meet the requirements of real-time vehicle control.

This chapter draws on the idea of penalty cost constraint to design an improved particle swarm algorithm with a penalty function, which serves as the fitness function of the particle swarm method together with the cost function in quadratic programming, so that multiple constraints can be transformed into unconstrained planning problems. And a random dynamic weight and compression factor are introduced to accelerate the algorithm's convergence and reduce the risk of getting trapped in a local optimum.

On this basis, a hot start is added, and based on the characteristics of the MPC optimal solution vector, $X = [\Delta u(k+1), \Delta u(k+2), \ldots, \Delta u(k+N_c), 0]$, the remaining solution vector control sequence, serves as the initial local optimal solution of the PSO method in the next cycle. Ultimately, the principle of heuristic search is used to guide the particles to reach the global optimal point quickly.

1) The penalty function is constructed in the following manner:

$$f_j(x_i) = Ax_i - b \quad (34)$$

$$P_j(x_i) = \max(0, f_j(x_i)) \quad (35)$$

$$F(x_i) = \sum_{j=1}^{n} P_j(x_i) \quad (36)$$

where $f_j(x_i)$ represents the performance value of the particle under the jth constraint; $P_j(x_i)$ is the penalty value of the particle under the jth constraint; and $F(x_i)$ is the penalty function of the particle under n total constraints.

1. The fitness function is constructed as follows:

$$S(x) = \frac{1}{2}x^T Hx + f^T x + F(x) \quad (37)$$

2. Introducing random dynamic weights:

$$w = w_{\min} + (w_{\max} - w_{\min}) * \exp\left(-\frac{t}{t_{\max}}\right) + \sigma * betarnd(p, q) \quad (38)$$

3. Adding compression factor:

$$V^{k+1} = \lambda(wV^k + c_1 r_1(P_{best} - X^k) + c_2 r_2(G_{best} - X^k)) \quad (39)$$

where $\lambda = \frac{2}{|2-\varphi-\sqrt{\varphi^2-4\varphi}|}, \varphi = c_1 + c_2 > 4$.

The optimization of the MPC's QP solver using the IPSO algorithm involves undertaking the following four steps:

First step: Initialize the population with a population size of 50; set the maximum number of iterations to 20; particle position: $X = [\Delta U(k)^T, \varepsilon]$, upper boundary: $X_{\max} = [\Delta\delta_{f_{\max}}, 10]$, lower boundary: $X_{\min} = [\Delta\delta_{f_{\min}}, 10]$; upper boundary of particle velocity$V_{\max} = [0.01, 0.01]$, lower boundary: $V_{\min} = [-0.01, -0.01]$; and$\sigma = 0.1, p = 1, q = 3$, $c_1 = c_2 = 2.1, w_{\min} = 0.4, w_{\max} = 0.9$.

Second step: Calculate the fitness value $S(x)$ under multiple constraints and update the individual best P_{best} and global best G_{best}.

Third step: Update the particle's speed and location according to the improved dynamic weights and compression factors.

$$\begin{cases} w = w_{\min} + (w_{\max} - w_{\min}) * \exp\left(-\frac{t}{t_{\max}}\right) + \sigma * betarnd(p, q) \\ V^{k+1} = wV^k + c_1 r_1(P_{best} - X^k) + c_2 r_2(G_{best} - X^k) \\ X^{k+1} = X^k + V^{k+1} \end{cases} \quad (40)$$

Fourth step: The process continues until the termination condition (maximum number of iterations) is satisfied, resulting in the output of the global optimal solutionδ_f.

6 | ONLINE ADAPTIVE SIMULATION

Although the offline adaptive controller can meet the tracking requirements in specific scenarios, once the scenario changes (e.g. vehicle speed, road adhesion coefficient), the offline adaptive controller must re-run the model until the optimal control parameters are found, which is a time-consuming process and not suitable for complex and changing driving scenarios (e.g.

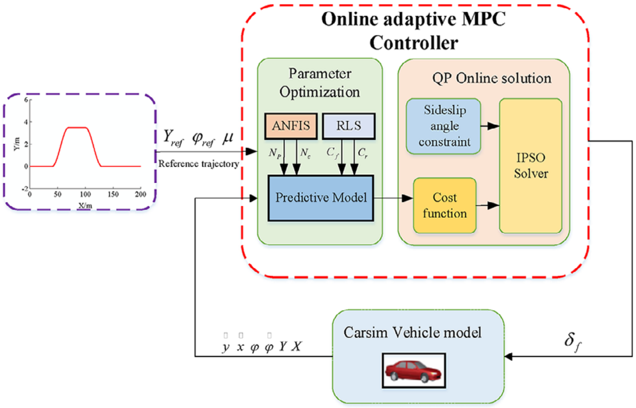

FIGURE 5 Block diagram of adaptive lateral control strategy.

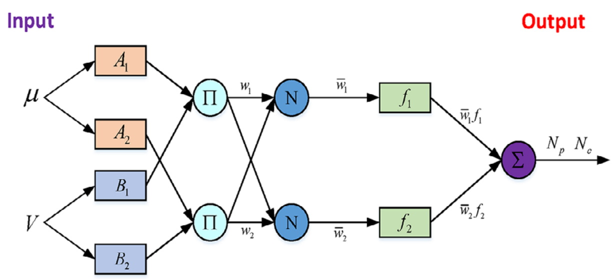

FIGURE 6 ANFIS network structure diagram.

variable vehicle speed, variable adhesion coefficient). Therefore, in this chapter, the offline adaptive controller is first used to find the optimal control parameters corresponding to varying vehicle velocities and road surface adhesion coefficients, and then the ANFIS is used to train this data set with inputs of road surface adhesion coefficients and vehicle velocities and outputs of prediction horizon and control horizon.

Finally, the trained ANFIS and cornering stiffness estimator are used as the upper layer parameter selection module, and the IPSO solver is used as the lower layer solution module to build the online adaptive lateral controller. The detailed diagram for the control strategy is presented in Figure 5.

ANFIS is a neural network incorporating fuzzy inference that introduces the learning ability of neural networks into fuzzy systems, greatly enhancing the inference ability of fuzzy systems. ANFIS has ready-to-use toolboxes in MATLAB, and the specific network architecture is presented in Figure 6.

The vehicle parameters and controller parameters are presented in Table 1 and 2, respectively. The road surface adhesion coefficient interval (0.3–0.9) is divided into equal intervals of 0.1, and the speed interval (10–120 km/h) is divided into equal speed intervals of 10 km/h. Multiple simulations were performed using an offline adaptive controller at various fixed vehicle speeds and road adhesion coefficients, and the optimal time horizon parameters (prediction horizon N_p and control horizon N_c) were recorded. Some of the data sets are shown in Tables 3 and 4.

TABLE 1 Vehicle parameter.

Symbol	Definition	Value (unit)
m	Vehicle mass	1723 kg
a	Distance from the front axle to the vehicle's centroid	1.232 m
b	Distance from the rear axle to the vehicle's centroid.	1.468 m
h	The height of the vehicle's centroid	0.51 m
I_z	Yaw mass moment of inertia	4175 kg·m^2
d	Track width	1.3 m
r	Wheel radius	0.302 m

TABLE 2 Controller parameter.

Symbol	Parameters	Value (unit)
T	Sampling time	0.02 s
Q	Error weights	Diag[100,50]
R	Control weights	100,000
ρ	Relaxation factor weights	1000
δ_f	Control quantity	−25° to 25°
$\Delta\delta_f$	Control increment	−0.47° to 0.47°
$\alpha_{\min}$	Minimum sideslip angle	−2°
$\alpha_{\max}$	Maximum sideslip angle	2°

TABLE 3 The optimal time horizon parameters for each vehicle speed at a coefficient of adhesion of 0.8.

Speed (km/h)	Predictive horizon	Control horizon
10	11	10
20	11	9
30	12	11
40	12	11
50	13	10
60	15	12
70	16	10
80	18	8
90	25	10
100	32	7
110	40	5
120	40	1

6.1 | Docked road conditions

To assess the tracking performance of the adaptive MPC controller on varying adhesion road surfaces, a test scenario is designed with a docked road surface, as illustrated in Figure 7a. The first half of the road features a high-adhesion surface ($\mu = 0.8$), while the second half comprises a low-adhesion sur-

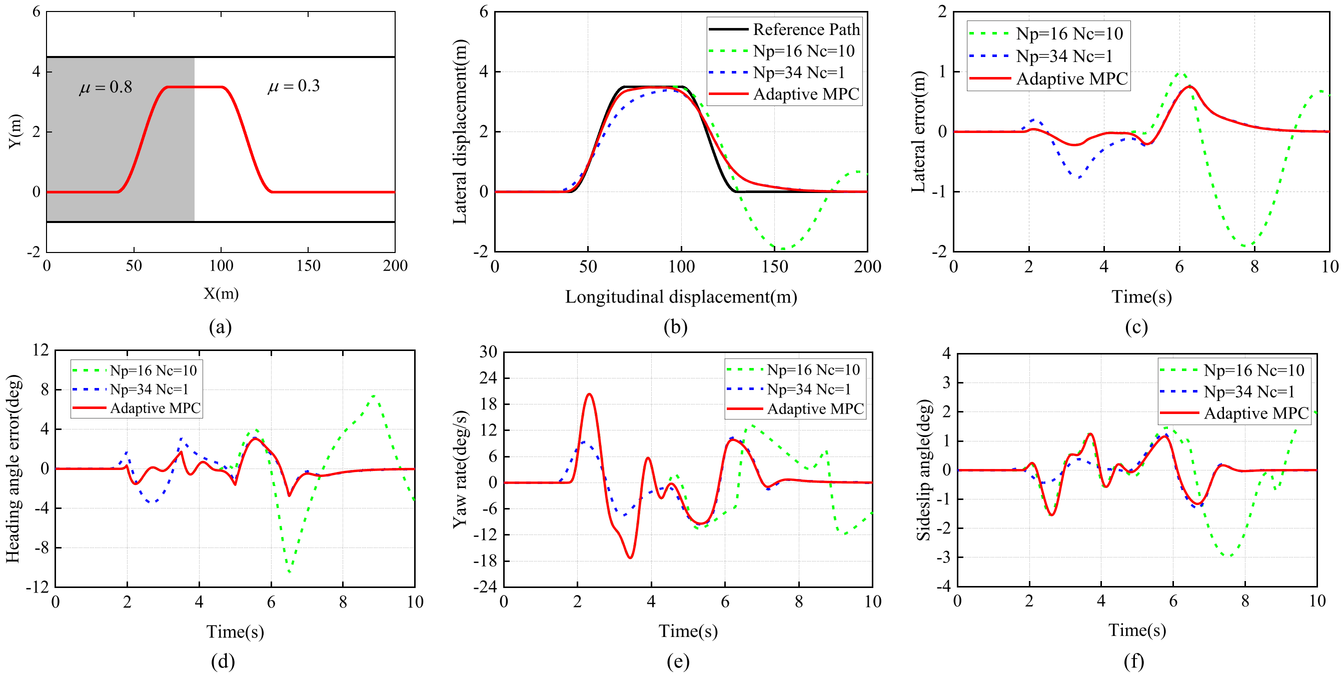

FIGURE 7 (a) Schematic diagram of docked road surface, where μ represents the road surface adhesion coefficient; simulation results of different controllers on a docked road surface: (b) tracking effect, (c) lateral error, (d) heading angle error, (e) yaw rate, (f) sideslip angle.

TABLE 4 the optimal time horizon parameters for various adhesion co-efficient at a speed of 100 km/h.

Road adhesion coefficient	Predictive horizon	Control horizon
0.3	40	1
0.4	38	1
0.5	34	2
0.6	31	2
0.7	30	3
0.8	28	5
0.9	25	7

face ($\mu = 0.3$). And the vehicle maintains a constant velocity of 72 km/h and tracks an avoidance overtaking trajectory.

From Figure 7b, it can be observed that when the controller parameters are selected as the optimal time horizon parameter ($N_P = 16, N_c = 10$) under the operating conditions of a constant velocity of 72 km/h and $\mu = 0.8$, although the controller can follow the desired trajectory well in the first half, when the road adhesion coefficient becomes 0.3 in the latter half, the controller parameters still prioritize tracking accuracy, resulting in dangerous rear axle sliding of the vehicle; when the controller parameters are selected as the optimal time horizon parameters ($N_P = 34, N_c = 1$) under the operating conditions of a constant velocity of 72 km/h and $\mu = 0.3$, the controller can complete the tracking task throughout, but on high adhesion roads, the tracking accuracy is low due to the controller parameters focusing on tracking stability; the adaptive MPC can adjust the controller parameters adaptively according to road adhesion and vehicle speed—time horizon parameters ($N_P = 16, N_c = 10$) are used in the first half and time horizon parameters ($N_P = 34, N_c = 1$) in the latter half, which can ensure that the controller focuses on tracking accuracy in the first half and tracking stability in the latter half—so that the adaptive MPC can not only enhance the vehicle's ability to follow the desired trajectory but also improve its driving stability. As shown in Figure 7c–f, the adaptive MPC controller demonstrates larger yaw rate as well as sideslip angle of the centre of gravity during the first half, resulting in better tracking accuracy with smaller lateral error and heading angle error. While in the second half, the adaptive MPC controller exhibits smaller yaw rate as well as sideslip angle of the centre of gravity, leading to larger lateral error but better stability of the controller compared to the other two controllers.

6.2 | Split road conditions

To assess the tracking capability of the adaptive controller under continuous variable adhesion road surfaces, the road surface is configured as a split road surface, as depicted in Figure 8a. The left lane represents a low-adhesion surface ($\mu = 0.3$), while the right lane represents a high-adhesion surface ($\mu = 0.8$). The vehicle tracks an avoidance overtaking trajectory while maintaining a constant velocity of 72 km/h.

Figure 8b illustrates that when the controller parameters are selected as the optimal time horizon parameters ($N_P = 16, N_c = 10$) under the working conditions with a constant velocity of 72 km/h and $\mu = 0.8$, although the controller can

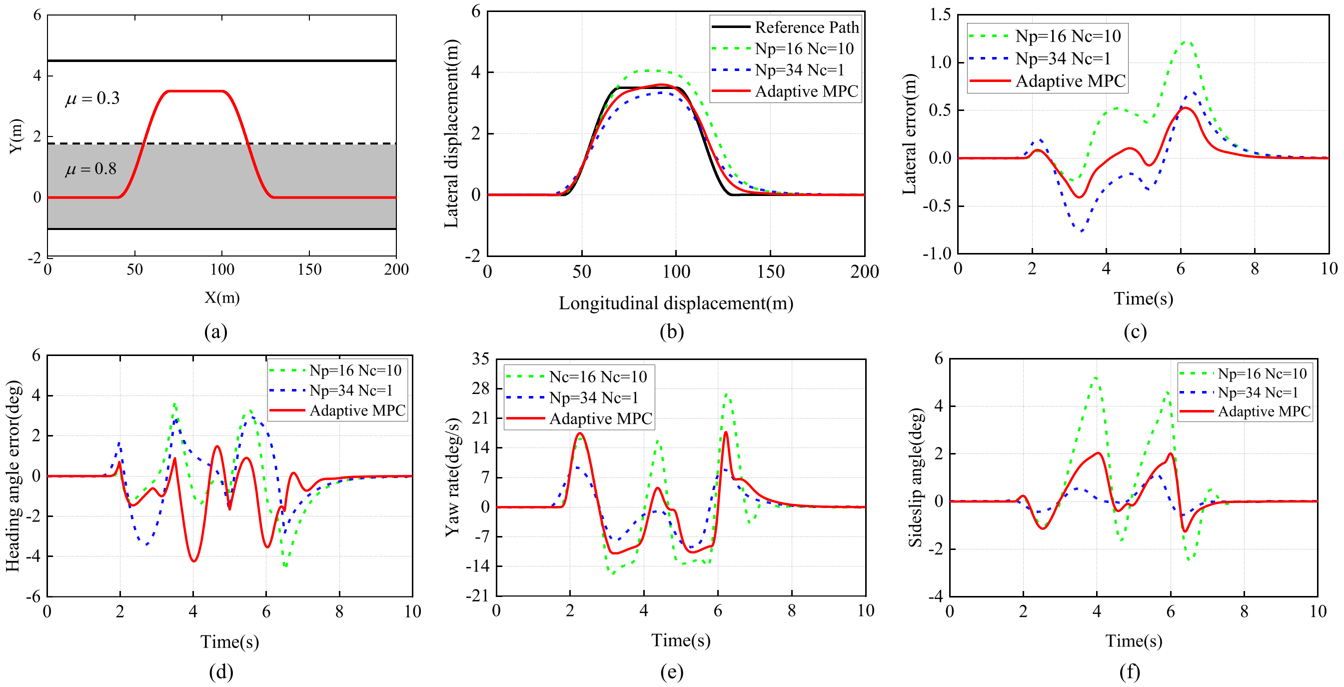

FIGURE 8 (a) Schematic diagram of split road surface; simulation results of different controllers on split road surface: (b) tracking effect; (c) lateral error; (d) heading angle error; (e) yaw rate; (f) sideslip angle.

accurately track the reference path in the right lane (high adhesion road), in the left lane with low adhesion road, the pursuit of tracking accuracy leads to the tire not providing enough lateral force to balance the steering required centripetal force and eventually deviates from the reference path; when the controller parameters are chosen as the optimal time horizon parameters ($N_P = 34, N_c = 1$) under the working conditions with a constant velocity of 72 km/h and $\mu = 0.3$, the controller is well stabilized, but its tracking accuracy is low; while the adaptive MPC can change the controller parameters in time to adjust the tracking accuracy and stability, its tracking effect is best compared with the other two fixed time horizon controllers. In Figure 8c,d, it is evident that the adaptive MPC has the smallest lateral error and heading angle error compared to the other two controllers with fixed time horizon. In Figure 8e,f, it is clear that the adaptive MPC is dominated by tracking accuracy on good road surfaces, and the yaw rate as well as vehicle sideslip angle exhibit relatively large values but remain within the dynamical constraints; the adaptive MPC is dominated by tracking stability on low adhesion roads, resulting in smaller yaw rate and vehicle sideslip angle.

6.3 | Variable speed conditions

In the previous simulation experiments, the speed was always set to a fixed value, which does not really reflect the controller's adaptability to speed. To assess the tracking

capability of the adaptive MPC controller under varying vehicle speeds, the velocity of the vehicle is ramped up from 20 to 96 km/h and $\mu = 0.85$. The velocity variation curve with respect to position is illustrated in Figure 9a.

In Figure 9b,c,d, it is evident that eight groups of fixed time horizon parameter MPC controllers and adaptive MPC were compared for tracking effectiveness. Except for the controllers with fixed time horizon parameters in ($N_P = 45, N_c = 1$), all the remaining seven controllers with fixed time horizon parameters failed to track. This is because with the continuous increase of vehicle speed, the controller needs a larger prediction horizon to grasp further path information and a smaller control horizon to limit the frequency of high-speed steering. Although this controller ($N_P = 45, N_c = 1$) can track the reference path stably, its focus on tracking stability throughout leads to sacrificing significant tracking accuracy at lower vehicle speeds, which is an inherent flaw of fixed time horizon parameter controllers. Only the adaptive MPC is constantly adjusting the time horizon parameters of the MPC based on vehicle speed information, utilizing a smaller prediction horizon and a larger control horizon at lower speeds to improve precision in tracking of the controller, and the prediction horizon is gradually increased and the control horizon is decreased as the speed continues to rise to ensure tracking stability under high-speed conditions. Therefore, during the whole variable speed tracking process, the adaptive MPC has the smallest tracking error, and no dangerous rear axle sideslip occurs. In Figure 9e,f, it is clear that with the increasing speed, the yaw rate as well as centre of mass sideslip angle curve of the other seven controllers with fixed time horizon parameters fluctuate greatly after about 8 s, causing a sharp decrease in stability of the vehicle; while the yaw rate as well as vehicle sideslip angle curve of the adaptive

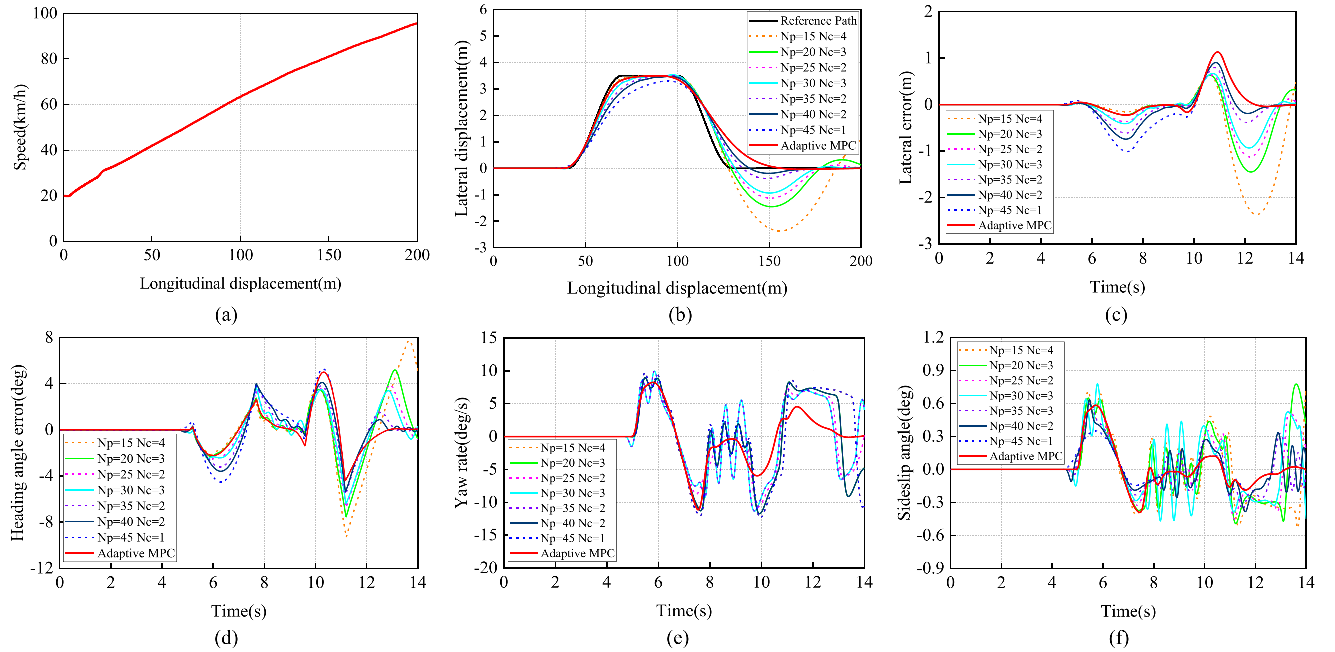

FIGURE 9 (a) Schematic diagram of variable vehicle speed; simulation results for different controllers at variable vehicle speeds: (b) tracking effect, (c) lateral error, (d) heading angle error, (e) yaw rate, (f) sideslip angle.

MPC change smoothly throughout the whole process, with a small amplitude of change, and the vehicle driving stability is good.

In summary, the efficiency and robustness of the designed adaptive MPC controller are fully verified through simulation experiments under three working conditions: docked road surface, split road surface, and variable vehicle speed. The accuracy and stability of path tracking are inherently contradictory and cannot be achieved simultaneously. However, the adaptive MPC takes into full account the information regarding velocity of the vehicle and road surface and make adaptive adjustments to its time horizon parameters so that it can achieve the different requirements of tracking accuracy and driving stability under different working conditions. Implementing such an adaptive MPC controller in practical automated driving systems requires powerful processors, typically multi-core CPUs and GPUs, to perform complex control algorithms in real-time. These processors should not only handle adaptive MPC but also manage other critical tasks such as perception, planning, and sensor fusion.

7 | CONCLUSIONS

The conventional MPC controller cannot adapt to the complex and changing tracking environment and have poor tracking effects, so the adaptive MPC controller was designed in this paper.

Firstly, a tire cornering stiffness estimator using the RLS method was developed to update the tire model parameters. Secondly, a flexible PSO algorithm is used to optimize the MPC controller's time horizon parameters offline under different road conditions and vehicle speeds, forming a dataset to train an ANFIS for online adaptive parameter tuning. Thirdly, to further enhance the precision of quadratic programming, an IPSO algorithm that can handle multi-constraint objective problems was designed. Finally, a collaborative simulation platform was constructed using MATLAB/Simulink and Carsim for online adaptive condition verification. The results indicate that: (1) The adaptive MPC controller exhibits minimal lateral and heading angle errors, demonstrating high tracking accuracy. (2) The yaw rate and sideslip angle of the adaptive MPC controller remain within dynamic constraints, displaying smooth and minimal variations throughout the entire course, resulting in excellent vehicle stability.

Future research should focus on accurate modelling of complex vehicle dynamics and real-time trajectory tracking, as well as effectively dealing with various constraints in dynamic traffic environments.

AUTHOR CONTRIBUTIONS

Xinyong Liu: Writing—original draft; methodology; review and editing. **Jian Ou**: Funding acquisition; supervision. **Dehai Yan**: Software; writing—review and editing. **Yong Zhang**: Project administration. **Guohong Deng**: Resources.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data is not provided.

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